KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5241 | 5 Credits: 4 | Program Core | Source-grounded

Medical Imaging Techniques

 To lay foundation of various imaging modalities in the medical field.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5241 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5241.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	5241
Course title	Medical Imaging Techniques
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To lay foundation of various imaging modalities in the medical field.
- □ To impart knowledge on the principle of operation and instrumentation of various
- imaging techniques and its applications.

Course outcomes

CO	Official outcome	Study level
CO1	and Fluoroscopy machine based on working 20 Understanding principle and instrumentation. Describe the process of image formation in	See official syllabus
CO2	ultrasound machine and list the applications of 11 Understanding ultrasound imaging. Identify the parts of the MRI machines and explain	See official syllabus
CO3	16 Understanding the role of each part. Explain nuclear imaging modalities and relate its	See official syllabus
CO4	11 Understanding applications in the medical field.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	machine based on working principle and instrumentation.	4	As specified
Module 2	the applications of ultrasound imaging.	4	As specified
Module 3	Identify the parts of the MRI machines and explain the role of each part.	4	As specified
Module 4	medical field.	4	As specified

MODULE 1

machine based on working principle and instrumentation.

This module is presented from the official syllabus scope for Medical Imaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: machine based on working principle and instrumentation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ **3 Understanding rays Illustrate the working of x-ray machine and**

3 Understanding rays Illustrate the working of x-ray machine and is an official topic in Module 1 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding rays Illustrate the working of x-ray machine and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding rays illustrate the working of x-ray machine and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding rays Illustrate the working of x-ray machine and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding rays Illustrate the working of x-ray machine and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▼ list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block

list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block is an official topic in Module 1 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the troubleshooting procedure of x-ray 5 understanding machine illustrate the working principle and block should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 8 Understanding diagram of CT machine Summarize the working of interventional

8 Understanding diagram of CT machine Summarize the working of interventional is an official topic in Module 1 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding diagram of CT machine Summarize the working of interventional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding diagram of ct machine summarize the working of interventional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 8 Understanding diagram of CT machine Summarize the working of interventional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 8 Understanding diagram of CT machine Summarize the working of interventional □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list

4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list is an official topic in Module 1 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine-Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding radiological equipment x-ray machine and ct machine definition of x- rays, properties of x-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, block diagram of x-ray machine. principle of digital radiography. computed tomography - basic principle of computed tomography - linear attenuation coefficient - ct number - concept of image reconstruction, block diagram of ct machine- scanning unit - ct generations - x-ray detectors(scintillation and xenon) - processing system-viewing system, interventional radiography – fluoroscopy - block diagram of c-arm, applications of fluoroscopy - angiography, angioplasty. case study : prepare a presentation on the troubleshooting procedure of an x-ray machine describe the process of image formation in ultrasound machine and list should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

the applications of ultrasound imaging.

This module is presented from the official syllabus scope for Medical Imaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: the applications of ultrasound imaging.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in

it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in is an official topic in Module 2 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, it with principle of operation of ultrasound 4 understanding scanners. illustrate the modes of scanning used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible].

▼ 3 Understanding ultrasound imaging

3 Understanding ultrasound imaging is an official topic in Module 2 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding ultrasound imaging using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding ultrasound imaging should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding ultrasound imaging means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding ultrasound imaging □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Describe the types of ultrasound scanners

Describe the types of ultrasound scanners is an official topic in Module 2 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the types of ultrasound scanners using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the types of ultrasound scanners should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the types of ultrasound scanners means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe the types of ultrasound scanners □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging.

List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging. is an official topic in Module 2 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound -

properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the applications of ultrasound scanning 1 remembering ultrasound machine ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - doppler effect. generation of ultrasound - piezoelectric effect, ultrasound transducer construction. modes of scanning - a-mode, b-mode & m-mode, block diagram of a-mode and b-mode scanners. types of scanners - linear scanner, sector scanner, compound scanners & real time scanners (concept only). applications of ultrasound - doppler ultrasound - echocardiography – echoencephalography - other applications in medical imaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real

Aspect	What to prepare
	time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the parts of the MRI machines and explain the role of each part.

This module is presented from the official syllabus scope for Medical Imaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the parts of the MRI machines and explain the role of each part.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Describe the principle of operation of MRI

Describe the principle of operation of MRI is an official topic in Module 3 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the principle of operation of MRI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the principle of operation of mri should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the principle of operation of MRI means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Describe the principle of operation of MRI [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine

Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine is an official topic in Module 3 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of relaxation times in mri 2 understanding illustrate the block diagram of mri machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the

6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the is an official topic in Module 3 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding and identify the parts of mri machine list the applications of mri and explain the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain nuclear imaging modalities and relate its applications in the

3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain nuclear imaging modalities and relate its applications in the is an official topic in Module 3 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain nuclear imaging modalities and relate its applications in the using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding principle of fmri mri machine magnetic resonance imaging - principle of operation - concept of magnetic dipole - precession - lamour frequency - concept of resonance- rf excitation - fid signal - longitudinal and transverse relaxation time - concept of image formation. block diagram of an mri machine - magnets used in mri - rf coils - gradient coils - pulse sequence (basic spin-echo) - functioning of each part. applications of mri - cardiology, neurology, orthopedics etc functional mri (fmri) - working principle. explain nuclear imaging modalities and relate its applications in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain

Aspect	What to prepare
	nuclear imaging modalities and relate its applications in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain nuclear imaging modalities and relate its applications in the □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the

Malayalam support notes.

MODULE 4

medical field.

This module is presented from the official syllabus scope for Medical Imaging Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: medical field.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding radioactive decay

3 Understanding radioactive decay is an official topic in Module 4 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding radioactive decay using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding radioactive decay should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding radioactive decay means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding radioactive decay □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block

Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block is an official topic in Module 4 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of gamma camera. 2 understanding describe the principle of operation and block should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Understanding diagram of SPECT scanner Describe the principle of operation and block

3 Understanding diagram of SPECT scanner Describe the principle of operation and block is an official topic in Module 4 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding diagram of SPECT scanner Describe the principle of operation and block using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding diagram of spect scanner describe the principle of operation and block should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding diagram of SPECT scanner Describe the principle of operation and block means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding diagram of SPECT scanner Describe the principle of operation and block □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation -block diagram- Application - isotopes used. Case study: Prepare a video on operation of a Gamma camera and radiopharmaceutical preparation. Text / Reference: T/R Book Title/Author RS Khandpur, Handbook of Biomedical Instrumentation- 3rd Edition, Tata T1 McGraw-Hill Pub. India, 2014 Steve Webb, Physics of Medical Imaging- 2nd Edition, Institute of Physics T2 Publishing, U K, 2008 Massey & Meredith ,Fundamentals Physics of radiology- J. Wright; 2nd edition - R1 1972 A C Kak, Principles of computerized tomographic imaging,1st edition, - Society R2 for Industrial and Applied Mathematics, Philadelphia, 2001 Govind B Chavhan, MRI Made Easy (for Beginners), 1st edition, Jaypee Brothers R3 Medical Publishers Pvt. Ltd, India, 2013 Online resources: Sl. No Website Link 1 https://www.youtube.com/watch?v=whp2pnCBs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=DSX0ygI0fZs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=IAWnRnqpVP0 6 https://www.youtube.com/watch?v=bXxqGWj2340 7 https://www.youtube.com/watch?v=NKVyHFk01lY https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students

3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation - Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation -block diagram- Application - isotopes used. Case study: Prepare a video on operation of a Gamma camera and

radiopharmaceutical preparation. Text / Reference: T/R Book Title/Author RS Khandpur, Handbook of Biomedical Instrumentation- 3rd Edition, Tata T1 McGraw-Hill Pub. India, 2014 Steve Webb, Physics of Medical Imaging- 2nd Edition, Institute of Physics T2 Publishing, U K, 2008 Massey & Meredith ,Fundamentals Physics of radiology- J. Wright; 2nd edition - R1 1972 A C Kak, Principles of computerized tomographic imaging,1st edition, - Society R2 for Industrial and Applied Mathematics, Philadelphia, 2001 Govind B Chavhan, MRI Made Easy (for Beginners), 1st edition, Jaypee Brothers R3 Medical Publishers Pvt. Ltd, India, 2013 Online resources: Sl. No Website Link 1 https://www.youtube.com/watch?v=whp2pnCBs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=DSX0ygI0fZs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=IAWnRnqpVP0 6 https://www.youtube.com/watch?v=bXxqGWj2340 7 https://www.youtube.com/watch?v=NKVyHFk01lY https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students is an official topic in Module 4 of Medical Imaging Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation -block diagram- Application - isotopes used. Case

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- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding diagram of pet scanner nuclear imaging concept of nuclear imaging- radioactive decay-half life period - properties of gamma rays. gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - applications of gamma camera - isotopes used, emission computed tomography- pet scanner - principle of operation - annihilation - block diagram - applications - isotopes used. spect scanners - principle of operation -block diagram- application - isotopes used. case study: prepare a video on operation of a gamma camera and radiopharmaceutical preparation. text / reference: t/r book title/author rs khandpur, handbook of biomedical instrumentation- 3rd edition, tata t1 mcgraw-hill pub. india, 2014 steve webb, physics of medical imaging- 2nd edition, institute of physics t2 publishing, u k, 2008 massey & meredith ,fundamentals physics of radiology- j. wright; 2nd edition - r1 1972 a c kak, principles of computerized tomographic imaging,1st edition, - society r2 for industrial and applied mathematics, philadelphia, 2001 govind b chavhan, mri made easy (for beginners), 1st edition, jaypee brothers r3 medical publishers pvt. ltd, india, 2013 online resources: sl. no website link 1 https://www.youtube.com/watch?v=whp2pncbs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=dsx0ygi0fzs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=iawnrnqpvp0 6 https://www.youtube.com/watch?v=bxxqgwj2340 7 https://www.youtube.com/watch?v=nkvyhfk01ly https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation -block diagram- Application - isotopes used. Case study: Prepare a video on operation of a Gamma camera and radiopharmaceutical preparation. Text / Reference: T/R Book Title/Author RS Khandpur, Handbook of Biomedical Instrumentation- 3rd Edition, Tata T1 McGraw-Hill Pub. India, 2014 Steve Webb, Physics of Medical Imaging- 2nd Edition, Institute of Physics T2 Publishing, U K, 2008 Massey & Meredith ,Fundamentals Physics of radiology- J. Wright; 2nd edition - R1 1972 A C Kak, Principles of computerized tomographic imaging,1st edition, - Society R2 for Industrial and Applied Mathematics, Philadelphia, 2001 Govind B Chavhan, MRI Made Easy (for Beginners), 1st edition, Jaypee Brothers R3 Medical Publishers Pvt. Ltd, India, 2013 Online resources: Sl. No Website Link 1 https://www.youtube.com/watch?v=whp2pnCBs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=DSX0ygI0fZs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=IAWnRnqpVP0 6 https://www.youtube.com/watch?v=bXxqGWj2340 7

Aspect	What to prepare
	https://www.youtube.com/watch?v=NKVyHFk01lY https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle-block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation -block diagram-Application - isotopes used. Case study: Prepare a video on operation of a Gamma camera and radiopharmaceutical preparation. Text / Reference: T/R Book Title/Author RS Khandpur, Handbook of Biomedical Instrumentation-3rd Edition, Tata T1 McGraw-Hill Pub. India, 2014 Steve Webb, Physics of Medical Imaging- 2nd Edition, Institute of Physics T2 Publishing, U K, 2008 Massey & Meredith ,Fundamentals Physics of radiology- J. Wright; 2nd edition - R1 1972 A C Kak, Principles of computerized tomographic imaging,1st edition, - Society R2 for Industrial and Applied Mathematics, Philadelphia, 2001 Govind B Chavhan, MRI Made Easy (for Beginners), 1st edition, Jaypee Brothers R3 Medical Publishers Pvt. Ltd, India, 2013 Online resources: Sl. No Website Link 1 https://www.youtube.com/watch?v=whp2pnCBs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=DSX0ygI0fZs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=IAWnRnqpVP0 6 https://www.youtube.com/watch?v=bXxqGWj2340 7 https://www.youtube.com/watch?v=NKVyHFk01lY https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 3 Understanding rays Illustrate the working of x-ray machine and. (3 marks)**

▶ **Define or explain list the troubleshooting procedure of X-ray 5 Understanding machine Illustrate the working principle and block. (3 marks)**

▶ **Define or explain 8 Understanding diagram of CT machine Summarize the working of interventional. (3 marks)**

▶ **Define or explain 4 Understanding radiological equipment X-ray machine and CT machine Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine Describe the process of image formation in ultrasound machine and list. (3 marks)**

Module 2

▶ **Define or explain it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in. (3 marks)**

▶ **Define or explain 3 Understanding ultrasound imaging. (3 marks)**

▶ **Define or explain Describe the types of ultrasound scanners. (3 marks)**

▶ **Define or explain List the applications of ultrasound scanning 1 Remembering Ultrasound Machine Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect. Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction. Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners. Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only). Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging.. (3 marks)**

Module 3

▶ **Define or explain Describe the principle of operation of MRI. (3 marks)**

▶ **Define or explain Explain the concept of relaxation times in MRI 2 Understanding Illustrate the block diagram of MRI machine. (3 marks)**

▶ **Define or explain 6 Understanding and identify the parts of MRI machine List the applications of MRI and explain the. (3 marks)**

▶ **Define or explain 3 Understanding principle of fMRI MRI machine Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation. Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part. Applications of MRI - cardiology, neurology, orthopedics etc Functional MRI (fMRI) - working principle. Explain nuclear imaging modalities and relate its applications in the. (3 marks)**

Module 4

▶ **Define or explain 3 Understanding radioactive decay. (3 marks)**

▶ **Define or explain Describe the working of Gamma camera. 2 Understanding Describe the principle of operation and block. (3 marks)**

▶ **Define or explain 3 Understanding diagram of SPECT scanner Describe the principle of operation and block. (3 marks)**

▶ **Define or explain 3 Understanding diagram of PET scanner Nuclear imaging Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays. Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used, Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used. SPECT scanners - Principle of operation - block diagram- Application - isotopes used. Case study: Prepare a video on operation of a Gamma camera and radiopharmaceutical preparation. Text / Reference: T/R Book Title/Author RS Khandpur, Handbook of Biomedical Instrumentation- 3rd Edition, Tata T1 McGraw-Hill Pub. India, 2014 Steve Webb, Physics of Medical Imaging- 2nd Edition, Institute of Physics T2 Publishing, U K, 2008 Massey & Meredith ,Fundamentals Physics of radiology- J. Wright; 2nd edition - R1 1972 A C Kak, Principles**

of computerized tomographic imaging,1st edition, - Society R2 for Industrial and Applied Mathematics, Philadelphia, 2001 Govind B Chavhan, MRI Made Easy (for Beginners), 1st edition, Jaypee Brothers R3 Medical Publishers Pvt. Ltd, India, 2013 Online resources: Sl. No Website Link 1 https://www.youtube.com/watch?v=whp2pnCBs6s 2 www.khanacademy.org 3 https://www.youtube.com/watch?v=DSX0ygI0fZs 4 https://www.wikihow.com 5 https://www.youtube.com/watch?v=IAWnRnqpVP0 6 https://www.youtube.com/watch?v=bXxqGWj2340 7 https://www.youtube.com/watch?v=NKVyHFk01lY https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a- 8 handbook-for-teachers-and-students. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding rays Illustrate the working of x-ray machine and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **it with principle of operation of ultrasound 4 Understanding scanners. Illustrate the modes of scanning used in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe the principle of operation of MRI**.

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **3 Understanding radioactive decay**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link https://www.youtube.com/watch?v=whp2pnCBs6s www.khanacademy.org https://www.youtube.com/watch?v=DSX0ygI0fZs https://www.wikihow.com https://www.youtube.com/watch?v=IAWnRnqpVP0 https://www.youtube.com/watch?v=bXxqGWj2340 https://www.youtube.com/watch?v=NKVyHFk01lY
- https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a-handbook-for-teachers-and-students

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5241. Verify institutional updates against the current official curriculum.